

ChE 454 MOLECULAR & MACROMOLECULAR ENGINEERING

Term: Spring 2021 Instructor: *Vivek Sharma*

The goal of the course to develop a comprehensive understanding of physicochemical principles underlying (or needed for) molecular and macromolecular engineering of the macroscopic optical, mechanical and transport properties of materials. We shall rely on the use of (a) the principles of statistical mechanics that will be introduced as part of the course, (b) a close reading of a few papers and selections from classic textbooks, and (c) curiosity-driven extra reading, primarily focused on polymers, liquid crystals and other soft materials. Familiarity with basic chemical engineering concepts (thermodynamics and transport phenomena typically taught at the undergraduate level) shall be assumed. We will discuss molecular and macromolecular basis of structure-property relationships, as well as how such relationships are deciphered in the macroscopic measurements that involve interaction with light with matter, or response to imposed mechanical stresses and other external fields. Though some overlap with an elective taught on *Fizzics and Interfacial Phenomena*, and core graduate courses on *Transport Phenomena* and *Microhydrodynamics* by the instructor can be expected, the course will be significantly different in content, coverage and emphasis.

Lectures on Monday and Wednesday, 1-2:15 PM.

Topics & Schedule

(A few topics will be added or altered based on the class participation, interest, and performance)

(PP: *Polymer Physics* by Rubinstein & Colby, IST: *Intermolecular and Surface Forces* by Israelachvili;

SMP: *Soft Matter Physics* by M. Doi; Hill's ST: *Statistical Thermodynamics*)

- **Molecular and Colloidal Reality of Matter.** Motivation for Molecular & Macromolecular Engineering of Materials including Complex Fluids. Historical overview: From Lucretius to Boltzmann, Perrin, Einstein and Staudinger. Molecular Reality. Colloidal, & Macromolecular Hypothesis. Brownian Motion. (Week 1) *Perrin's Nobel Lecture. Chandrashekar's 1943 Review.*
- **Hard vs. Soft Matter:** Elasticity and Viscosity. Order, Dislocations, Defects, and Microstructure. Crystal Structure. Liquid Crystals and Quasi-crystals. Pair-Correlation functions and Structure Factor. Light Scattering. (Week 1-2)
- **Macromolecular Reality of Matter:** Random Walks & Gaussian Chains. Kuhn chain, Persistence length and Entropic Elasticity. Size and Shape Characterization (using Sedimentation, Viscosity, Scattering, etc.). (Week 2-3) *Staudinger's Nobel Lecture. Einstein's 1905 paper. Flory's Nobel Lecture. PP-Ch.1-2. ZG Wang's Perspective 2017.*
- **Molecular and Macromolecular Thermodynamics:** Review of Solution Thermodynamics, Bragg-Williams and Flory-Huggins Theory. Real Polymers & Excluded Volume Interactions. Gelation. Phase Separation: Nucleation & Growth vs. Spinodal Decomposition. Block copolymer Phase Behavior (Week 4-5) *PP-Ch3-5. SMP Ch2. Balsara's Perspective 2017. Bates and Bates' Perspective 2017.*
- **Molecular and Macromolecular Dynamics:** Fluctuation-Dissipation Theorem. Stokes Drag, Stokeslet and Hydrodynamic Interactions. Bead-Spring models (Rouse/Zimm). Fluctuations and Blob models for Semi-dilute Solutions. Entangled Polymer Dynamics. (Week 6-7) *PP-Ch 8-9. Bead-Spring Model papers by Rouse (1953) and Zimm (1956). Schroeder's review (2018). Nagele's The Physics of Colloidal Soft Matter*
- **Classical Statistical Mechanics:** Microcanonical, canonical and grand canonical ensembles. Ideal Gas & Ideal Solution. Virial Expansion. Phase Transitions and Critical Phenomena. Curie Law. Ising Model. (Week 8) *Selections from Hill's ST.*
- **Surface Forces and Thermodynamics of Interfaces:** Gibbs and Langmuir adsorption equations. Charged interfaces and electrostatic double layer theory. Debye-Hückel approximation. Disjoining pressure. Dispersion forces. Non-DLVO forces. (Week 9) *Bergeron's review. ISF-selections. Nagele's The Physics of Colloidal Soft Matter*
- **Proteins and Polyelectrolytes:** Polyelectrolytes and Coacervation. Dynamics of Polyelectrolytes. Protein Structure and Dynamics. Protein Unfolding. Plant-based Proteins. (Week 10-11) *Dill's Review. SMP-Ch10. Muthukumar's Perspective 2017. Dobrynin's Book Chapter 2012. McClements' reviews of plant-based milks and food emulsions.*
- **Surfactants, Lipids and Liquid Crystals:** Self-Assembly. Micellization. Detergency. Solubilization. Digestion. Lyotropic and Thermotropic Liquid Crystals. Frank Elasticity in Liquid Crystals. Curvature Elasticity. (Week 11-12) *ISF-selections. SMP-Ch4-5. Nagarajan's review on micelles. Onsager's 1949 paper on rods.*
- **Molecular & Macromolecular Engineering of Complex Fluids:** Applications in Energy, Water, Medicine, Cosmetics, Coatings, Adhesives or Bio-inspired Materials (Week 13)
- **Special Topics:** Viruses, mRNA Vaccines, "Coronavirus" and Pharmaceuticals. (Week 14)
- **Individual projects, presentations and term papers as Finals.**